

Kai-Hua Wang

☎ (+886) 968-831-106

✉ kaihuawang.work@gmail.com

🌐 Kai-Hua Wang

🌐 Personal Website

Education

Carnegie Mellon University (CMU)

M.S. in Mechanical Engineering

Pittsburgh, PA

Expected: May 2028

- Focus Area: Robotics, Autonomous Systems, and Controls

National Taiwan University (NTU)

B.S. in Mechanical Engineering

Taipei, Taiwan

Sep 2021 – Jan 2026

- Overall GPA: 3.79 / 4.3, Last 60 credits: 4.12 / 4.3

Technical Skills

Programming Languages

C++, Python, C, MATLAB

Robotics & Controls

SLAM, Vehicle Dynamics, State Estimation, Controls, Sensor Fusion

Frameworks & Libraries

ROS/ROS2, PyTorch, OpenCV

Engineering Tools

Simulink, SolidWorks, COMSOL, STAR-CCM+

Languages

English (Proficient), Mandarin (Native)

Work Experience

Garmin Ltd.

New Taipei City, Taiwan

Manufacturing Engineer Intern, New Product Introduction (NPI)

Jul 2024 – Jan 2025

- **Partnered** with cross-functional electrical engineering teams to optimize automated testing equipment and calibrate inductive button parameters, **reducing the retest rate by 17.6%** and decreasing cycle time.
- **Designed and implemented** manufacturing workflows and optimized assembly fixture specifications using **Therblig analysis** and **PFMEA**, successfully transitioning the product to mass production.
- **Standardized** assembly processes and formulated fixture design parameters (including material selection and step-by-step procedures) to consistently achieve **IPX7 waterproof standards**.

Research Experience

System Optimization Laboratory, NTU; Advisor: Kuei-Yuan Chan

Sep 2024 – Jan 2026

Autonomous Robotic Straddle Carrier System Development

- **Designed and manufactured** a 1:12 scaled **autonomous straddle carrier AMR** with a motorized hoist, establishing a **Hardware-in-the-Loop (HIL)** testbed to validate perception and control prior to deployment.
- **Developed a redundant localization framework** by fusing **LiDAR SLAM** and **stereo-camera tracking** within the ROS2 environment, achieving robust state estimation and cross-validation under dynamic scenarios.
- **Engineered a hardware-level safety layer** by implementing an independent safety circuit with a dedicated power supply, **mitigating single-point failures** in dual-localization systems during critical edge-case testing.

Foxtron EV Autonomous Driving Control System Development (Exclusive Partner)

- **Pioneered** the vehicle control integration with Foxtron, developing a **deep-learning-based stereo-vision perception pipeline** and a high-fidelity 14-DOF vehicle dynamics model in **Simulink**.
- **Integrated** the comprehensive perception-control architecture within the **CARLA simulator**, successfully validating autonomous cone-lane navigation via a model predictive control (MPC) prior to vehicle deployment.

Smart and Advanced Manufacturing Laboratory, NTU; Advisor: Dian-Ru Li

Oct 2022 – Jan 2026

Biomechanical Mechanism Design & Verification (with Taipei Medical University)

- **Engineered** a novel double-lumen catheter aid to resolve clinical constraints identified by **Taipei Medical University** physicians, conducting user-centric research and human factors design in **SolidWorks**.
- **Formulated and executed Design of Experiments (DOE)** to translate clinical pain points into engineering parameters, successfully facilitating doctor-led validation for journal submission.

Formula SAE competition, Head of Aerodynamics, NTU Racing

Jul 2022 – Jul 2023

- **Led a 7-person aerodynamics team** to design high-performance aero kits, utilizing **COMSOL** and **MATLAB** to maximize lift-to-drag ratios and deliver **229.8 N of total downforce** at 11 m/s.
- **Advanced composite manufacturing workflows** by introducing Resin Transfer Molding (RTM), optimizing the strength-to-weight ratio to **reduce aero kit weight by over 25%** versus the prior generation.
- **Spearheaded vehicle aerodynamic testing** and cross-functional tuning, capturing **First Place overall** at the Formula Student Taiwan (FST) competition in 2023.